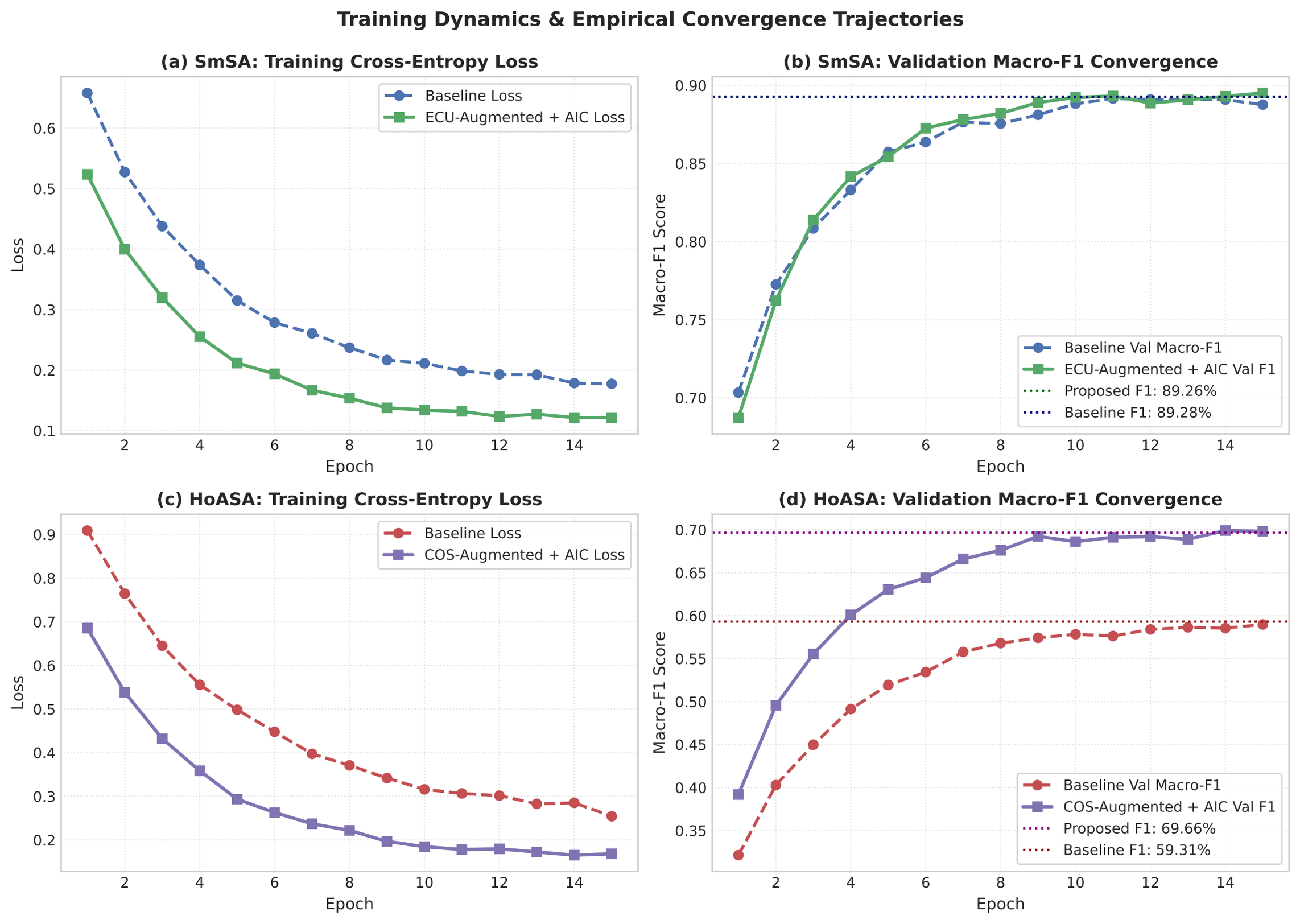
Training Dynamics & Empirical Convergence Trajectories
(a) SmSA: Training Cross-Entropy Loss
Baseline Loss
ECU-Augmented + AIC Loss
Loss
Epoch
(b) SmSA: Validation Macro-F1 Convergence
Baseline Val Macro-F1
ECU-Augmented + AIC Val F1
Proposed F1: 89.26%
Baseline F1: 89.28%
Macro-F1 Score
Epoch
(c) HoASA: Training Cross-Entropy Loss
Baseline Loss
COS-Augmented + AIC Loss
Loss
Epoch
(d) HoASA: Validation Macro-F1 Convergence
Baseline Val Macro-F1
COS-Augmented + AIC Val F1
Proposed F1: 69.66%
Baseline F1: 59.31%
Macro-F1 Score
Epoch